2.1.3 1D parallel dynamics

This subsection details attempts to treat various one-dimensional flows using finite elements (the `Firedrake` finite-element package is used). The treatment involves finding an analytic steady-state solution and then checking that numerical implementations recover said solution when evolved in time. It will be seen that an acceptable analytic solution only exists for certain choices of boundary condition. The equations will be presented in steady-state form only. The motivation for this work is to clarify when outflow solutions exist for a given set of equations and boundary conditions, in order to ease the implementation of the fully three-dimensional problem.

A domain spanning $[-1, 1]$ is used in all cases.

Generically n is density, u is velocity, T is temperature. Note that T is constant in the isothermal cases. The parameter τ is the ratio of ion to electron masses (c.2000 in the case of hydrogen, though it is common prctice to use lower values in order to ease the numerics).

`Firedrake` scripts referenced in this section can be accessed at [26].

Isothermal, single fluid (electrons only)

$$\begin{aligned} (nu)' &= \bar{n} \\ uu' &= -\tau T \frac{n'}{n}. \end{aligned} \tag{25}$$

The solution for u is

$$ue^{-\frac{u^2}{2\tau T}} = \alpha \bar{n} x, \tag{26}$$

where α is another constant of integration, which can be determined by requiring u to take particular outflow values $\pm u_0$ on the boundaries, thus $\alpha \bar{n} = u_0 e^{-\frac{u_0^2}{2\tau T}}$.

The equation above can be inverted using the Lambert W function, which is the solution $\mathrm{W}(z)$ of $\mathrm{W}e^{\mathrm{W}} = z$, and is computed numerically using *mpmath* [27]. In these terms the solution is

$$u = \pm \sqrt{-\tau T \, \mathrm{W}\left(-\frac{\alpha^2 \bar{n}^2 x^2}{\tau T}\right)}. \tag{27}$$

From this result it is possible to see that there is no regular solution if the outflow velocity is greater than $\sqrt{\tau T}$. Note the two real branches of the Lambert W function and consider its graph - the regular branch cannot return a value of magnitude greater than unity for a real negative argument (see Figure 5). If the solver is run with a Dirichlet condition enforcing an outflow velocity of greater magnitude than $\sqrt{\tau T}$ then there are obvious pathologies in the obtained steady-state solution e.g. discontinuities in the gradient. Note that regular subsonic outflows appear to be admissible in this

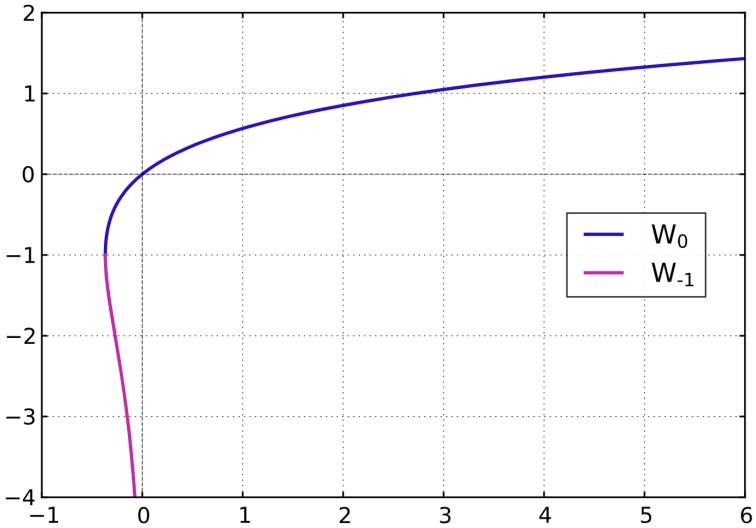

Figure 5: The two real branches W_0, W_{-1} of the Lambert function (the x-axis is the independent variable and the y-axis the dependent); only W_0 (blue curve) represents solutions that are regular at the origin, but it can be seen that this branch cannot return magnitude greater than unity for negative argument. Figure taken from [28].

case (nothing in the analytic solution appears to rule them out). Note that the solver can be run with a Dirichlet condition on the outflow velocity but the author typically uses a standard outflow condition in which the outflow speed is set to sonic values by the solver.

A cautionary note in running the analytically-forbidden cases numerically is always to be aware that the numerics contains additional effects such as numerical viscosity from the upwind DG method that means that expected singularities and discontinuities may not appear.

The time-dependent version of the system in Eqs.25 was simulated using `Firedrake`- see Figure 7. Parameter values of $\bar{n} = T = \tau = 1.0$ were used. The computational domain is spanned by 100 order-1 Lagrange elements, which are DG for density n and CG for velocity u. An implicit Gauss-

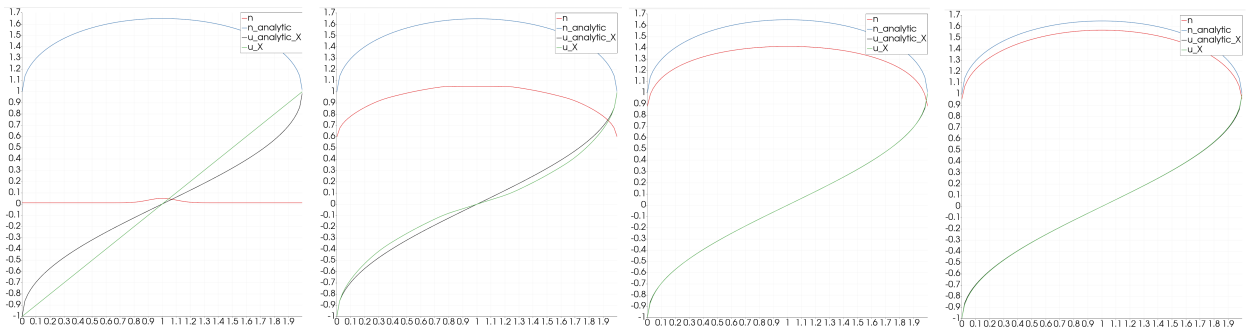

Figure 6: Evolution of the 1D flow from an initial Gaussian density bump / linear velocity condition toward the analytic solution given in the text, so the sonic outflow case. The numerical flow has $\bar{n} = 1.0$ and the time interval between frames is 1.5 time units.

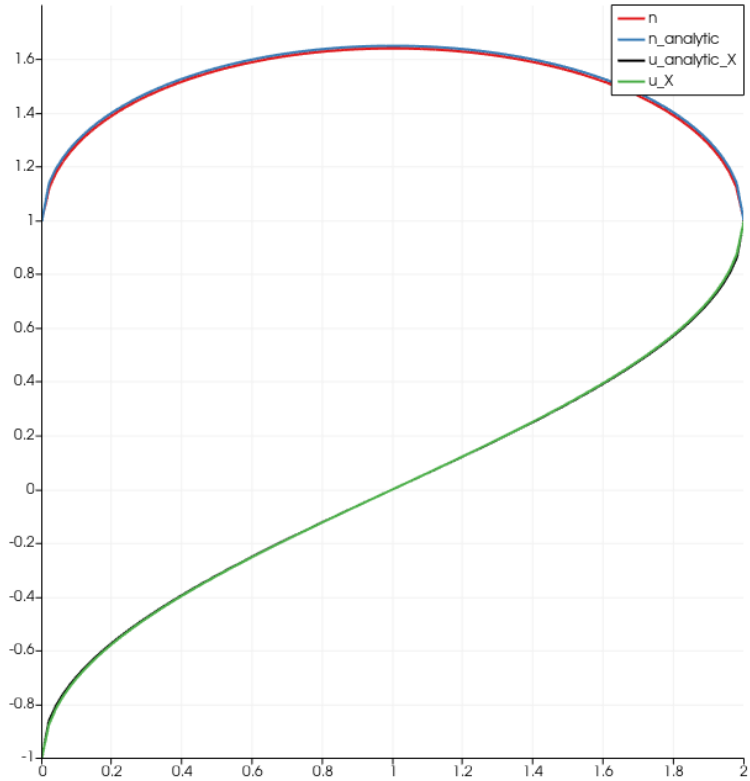

Figure 7: Visualization of n and u from evolving numerically Eqs.25 to the steady state. The curves can be seen to overlap near-precisely with the analytic solution described in the text.

Legendre time-stepper from *Irksome* was used.

Script: `RR_LAPD_1D.py`.

Isothermal, two fluids (ions and electrons)

The equations are, letting u_i be the ion velocity and u the electron velocity:

$$\begin{aligned} (nu)' &= \bar{n} \\ u_i u_i' &= -T\frac{n'}{n} \\ uu' &= -\tau T\frac{n'}{n}. \end{aligned} \tag{28}$$

The equations for n and u are independent of the ion velocity and have the solution as in the one-fluid example.

The equation for u_i is solved, again by directly integrating it, to (taking care not to confuse u and u_i)

$$ue^{-\frac{u_i^2}{2T}} = \beta\bar{n}x. \tag{29}$$

β is specified by the electron outflow velocity u_0 and the ion outflow velocity $u_{i,0}$ to $\beta\bar{n} = u_0 e^{-\frac{u_{i,0}^2}{2T}}$.

We specify that the ion outflow is sonic ($\pm\sqrt{T}$). This gives $\beta = \frac{\sqrt{\tau T}e^{-\frac{1}{2}}}{\bar{n}}$ in the case where the electron outflow velocity is $\sqrt{\tau T}$. The solution for u_i is

$$u_i = \pm\sqrt{2T\ln\frac{u}{\beta\bar{n}x}}. \tag{30}$$

Note that the boundary value of n can be deduced from the solution e.g. $\beta n = e^{-\frac{u_i^2}{2T}}$ so this does not need to be specified explicitly in the numerics.

It will now be shown that there is a good solution (free of singularities / discontinuities) for ion outflow velocity $\sqrt{T}$ only if the electron outflow velocity is exactly $\sqrt{\tau T}$.

First consider the overspeed case, $u_0 > \sqrt{\tau T}$. In this case regular solutions are impossible because the regular branch of the Lambert function cannot return a value of magnitude greater than unity for real negative argument. This means there are no regular solutions in this case (there are solutions from the -1 branch of the Lambert functions, but they have a logarithmic singularity at the domain centre). This is as in the one-fluid case.

Now consider the underspeed case, $u_0 < \sqrt{\tau T}$. Now in this case the requirement that the ion solution be continuous at the origin gives for the integration constants that $\alpha = \beta$, which is

$$u_0^2 = \tau u_{i,0}^2. \tag{31}$$

If this condition is not met then there is a discontinuity in the ion velocity at $x = 0$ of magnitude $2\sqrt{u_{i,0}^2 - \frac{u_0^2}{\tau}}$. Note this additional condition, which will be called the Bohm condition, arose from the presence of the ions and so it did not appear in the preceding examples.

The system in Eqs.29 was simulated using `Firedrake`- see Figure 8. Parameter values of $\bar{n} = T = 1.0$ and $\tau = 10.0$ were used. The computational domain is spanned by 100 order-1 Lagrange elements, which are DG for density n and CG for electron velocity u and ion velocity u_i. An implicit Gauss-Legendre time-stepper from *Irksome* was used.

Script: `RR_LAPD_1D_two_fluid.py`.

Isothermal, two fluids (ions and electrons) including electron drag

Including the parallel current into the equations in the preceding section, the steady-state equations, with ν a thermal conductivity, are

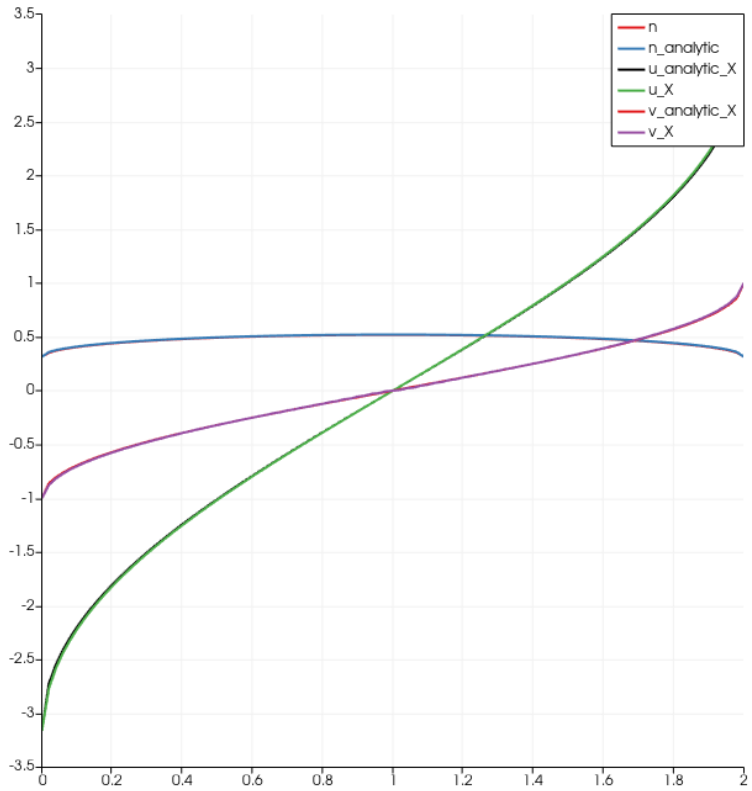

Figure 8: Visualization of n and u from evolving numerically Eqs.29 to the steady state. The curves can be seen to overlap near-precisely with the analytic solution described in the text.

$$
\begin{aligned}
(nu)' &= \bar{n} \\
u_i u_i' &= -T\frac{n'}{n} \\
uu' &= -\tau T\frac{n'}{n} + \tau\nu n(u_i - u).
\end{aligned} \tag{32}
$$

This system can be solved provided that the u_i term in the current is set to zero. Substituting for nu in the momentum equation gives

$$
uu' = -\tau T\frac{n'}{n} - \tau\nu\bar{n}x \tag{33}
$$

which integrates to

$$
ue^{-\frac{u^2}{2\tau T}} = \alpha\bar{n}xe^{\frac{\nu\bar{n}x^2}{2T}} \tag{34}
$$

with solution

$$
u = \pm\sqrt{-\tau T\ \mathrm{W}\left(-\frac{\alpha^2\bar{n}^2x^2e^{\frac{\nu\bar{n}x^2}{T}}}{\tau T}\right)}. \tag{35}
$$

Now setting the electron outflow speed to be u_0 one has $\alpha\bar{n} = u_0e^{-\frac{u_0^2+\tau\nu\bar{n}}{2\tau T}}$.

Note the now-familiar fact that there are no regular solutions with $u_0^2 > \tau T$ because the zero branch of the Lambert function cannot return a value of greater magnitude than unity for negative argument.

n and u_i can be obtained as above i.e. $n = \frac{\bar{n}x}{u}$ and $u_i = \pm\sqrt{2T\ln\frac{u}{\beta\bar{n}x}}$ wh. β is as before a constant of integration.

It is possible to show that this time there is a modifed Bohm condition. The ion outflow speed $u_{i,0}$ is set to be sonic, $u_{i,0} = \sqrt{T}$. Then the condition for continuity of the solution at $x = 0$ gives the condition on the integration constants $\alpha = \beta$. This gives the electron Bohm condition as

$$
u_0^2 = \tau\left(u_{i,0}^2 - \nu\bar{n}\right). \tag{36}
$$

For zero electron drag $\nu = 0$ the criterion is as above (Eq.31). For $\nu > 0$ the fact that electron momentum is absorbed means that the electron outflow speed is decreased over the no-drag case. It is seen that increased drag leads to enhanced density in the domain due to the decreased outflow.

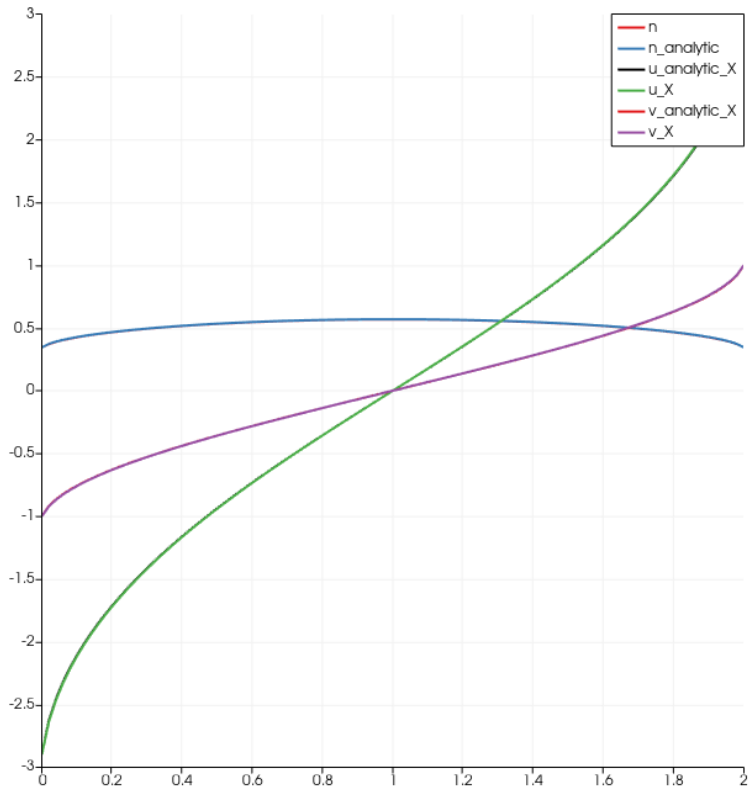

Figure 9: Visualization of n and u from evolving numerically Eqs.29 to the steady state. The curves can be seen to overlap near-precisely with the analytic solution described in the text. (Note that the ion speeds are labelled v in the plot.)

The system with steady state corresponding to Eqs.33 was simulated using `Firedrake`- see Figure 9. Parameter values of $\bar{n} = T = 1.0$ and $\tau = 10.0$ were used, along with the drag parameter $\nu = 0.16$. The computational domain is spanned by 100 order-1 Lagrange elements, which are DG for density n and CG for electron velocity u and ion velocity u_i. An implicit Gauss-Legendre time-stepper from *Irksome* was used. It is important to note that a Dirichlet condition on the outflow speed must be imposed in this case otherwise the outflow speed will not be consistent with the condition for a connected ion velocity profile at the origin for nonzero drag.

Script: `RR_LAPD_1D_two_fluid_and_drag.py`.

Non-isothermal, single-fluid (electrons only), equations in conservative form

We consider single-fluid (electrons only). The equations are modified slightly in order to make them more tractable and also easier to implement in `Nektar++`, which requires a conservative form - the temperature equation is modified from its original form $uT' = -\frac{2}{3}Tu' + \bar{T}$ (see [20] for the original 3D version).

The steady state satisfies, for C a dimensionless constant,

$$
\begin{aligned}
(nu)' &= \bar{n} \\
nuu' &= -\tau(nT)' \\
C(uT)' &= \left(\bar{T} = \bar{n}\right).
\end{aligned}
\tag{37}
$$

Clearly the steady-state n and T equations solve to $nu = \bar{n}x$ and $CTu = \bar{n}x$.

It is not difficult to integrate and solve for u, finding the cubic equation

$$
u^3 = Au - \frac{4\tau\bar{n}}{C}x. \tag{38}
$$

The integration constant A must be fixed by the boundary conditions of free outflow. To clarify these, the sound speed for this non-isothermal case must be evaluated.

The outflow speed can be determined as follows (in which it is assumed that $C = \tau = \bar{n} = 1$). Linearize the equations (setting $n = T$) and throw away derivatives of the base flow n_0, u_0 on the grounds that the wave is in the high-frequency limit. This gives

$$
\begin{aligned}
\dot{n} &= -n_0 u' - n' u_0 \\
\dot{u} &= -u_0 u' - 2\tau n'.
\end{aligned}
\tag{39}
$$

Assuming a harmonic wave one obtains the equation (eliminate n)

$$
\omega^2 + 2u_0\omega k + u_0^2 k^2 = 2\tau k^2 n_0 \tag{40}
$$

and use that $n_0 u_0 = \bar{n}$ at the endpoints, giving

$$
\frac{\omega^2}{k^2} + 2u_0\frac{\omega}{k} + u_0^2 = \frac{2}{u_0}. \tag{41}
$$

Now set $\frac{\omega}{k} = u_0$ (sonic outflow condition) to find simply

$$
u_0^3 = \frac{1}{2} \tag{42}
$$

i.e. the outflow speed is $u_0 = 2^{-\frac{1}{3}}$. This is observed in the numerical solution done with the usual weakly-imposed free outflow condition. Note if C is left general then the result is $u_0 = (2C)^{-\frac{1}{3}}$.

Now the cubic

$$u^3 = 3pu - 2q \tag{43}$$

has solution

$$u = \sqrt[3]{-q + \sqrt{q^2 - p^3}} + \sqrt[3]{-q - \sqrt{q^2 - p^3}} \tag{44}$$

and the discriminant can be taken as

$$\Delta = q^2 - p^3. \tag{45}$$

From the discriminant is seems that there are three distinct real roots and it is tricky to make the analytic formula give the correct branch, so a simple Newton-Raphson solve was used in order to plot the analytic solution (initial guess at zero; ten Newton iterations were found to be more than sufficient).

Note the solution in the $C = 2$, $\tau = \bar{n} = 1$ case plotted in the figure is explicitly

$$u = \sqrt[3]{-2x + \sqrt{4x^2 - \frac{27}{4}}} + \sqrt[3]{-2x - \sqrt{4x^2 - \frac{27}{4}}} \tag{46}$$

with $n = 2T$ and each specified by the other equations e.g. $nu = \bar{n}x$.

There does not seem, from this analytic solution, to be any problem with underspeed or overspeed solutions (where the outflow speed is respectively less than / greater than the speed of sound). At least for the parameter choice $C = \tau = \bar{n} = 1$ the inner solution branch always extends to $x = \pm 1$, as can be demonstrated by examining the curve $4x = Au - u^3$ and finding that the minimum x is precisely unity, for $u_0 = 2^{\frac{1}{3}}$ (which gives an infinite gradient at the outflow points). This conclusion seems to survive in the general case $(C, \tau, \bar{n})$.

The system in Eqs.38 was simulated using `Firedrake`- see Fig.38. Parameter values of $\bar{n} = T = \tau = 1.0$ and $C = 2.0$ were used. The computational domain is spanned by 100 order-1 Lagrange elements, which are DG for density n and CG for electron velocity u and ion velocity u_i. An implicit Gauss-Legendre time-stepper from *Irksome* was used.

Script: `RR_LAPD_1D_T_conservative_v2.py`

2.1.4 Challenges of implementing the full 3-D model

While the progress in modelling the parallel dynamics of the LAPD is encouraging, there remain a number of challenges to implementing a full 3-D model in `Nektar++`. One issue arises from the desire to use Discontinuous Galerkin discretisations to promote scalability in our solvers. Whilst many examples of DG solvers have been built using `Nektar`, they have tended to focus on equation systems which can straightforwardly be written in a conservative form. Indeed, this has been the case for the equations solved by all of the NEPTUNE proxyapps written up to this point that

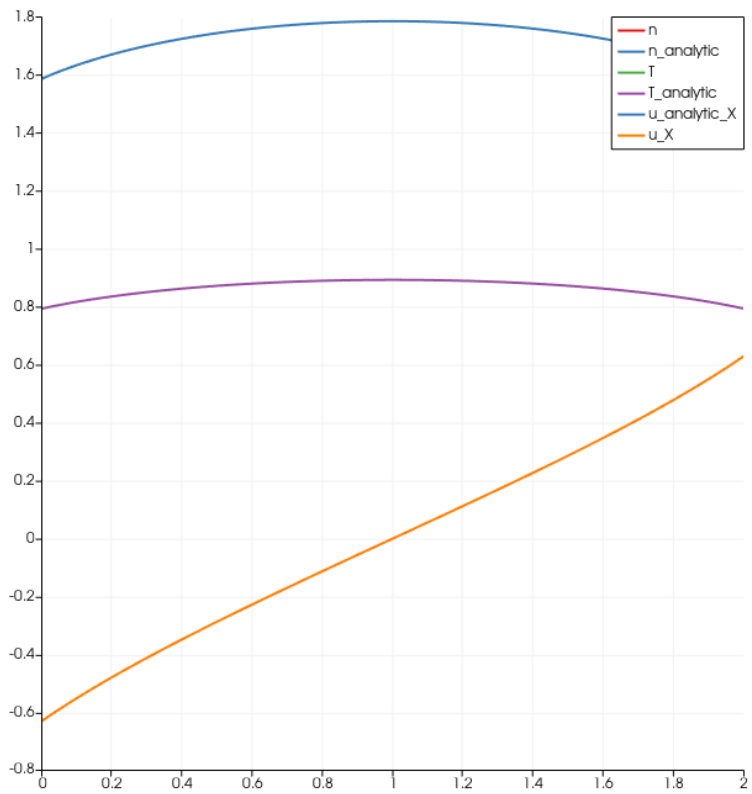

Figure 10: Visualization of n, u, and T from evolving numerically Eqs.33 to the steady state. The curves can be seen to overlap near-precisely with the analytic solution described in the text. Note that the outflow speed correctly corresponds to $4^{-\frac{1}{3}} \approx 0.63$, the same as the sound speed (which is $\sqrt{T}$ since here $\tau = 1$).